

Reproducing VISAtlas

An Image-Based Exploration and Query System for Large Visualization Collections

September 22, 2026

“The visualization collections are inherently diverse in many perspectives: from different data sources, by different research groups, and in different manners.”

“Existing systems for visualization images ... are mainly based on extrinsic attributes including author names and publication years.”

“More importantly, attribute-based systems neglect the intrinsic property, i.e., visual appearance of visualizations that is also perceived by humans.”

Although visual appearance is directly perceived by humans, it has often been overlooked as an analyzable property of visualization collections.

Y. Ye, R. Huang, W. Zeng. VISAtlas. IEEE TVCG 30(7):3224–3240, 2024.

“Which visualization collection shall be used?

Are there any similar visualizations in the collection?

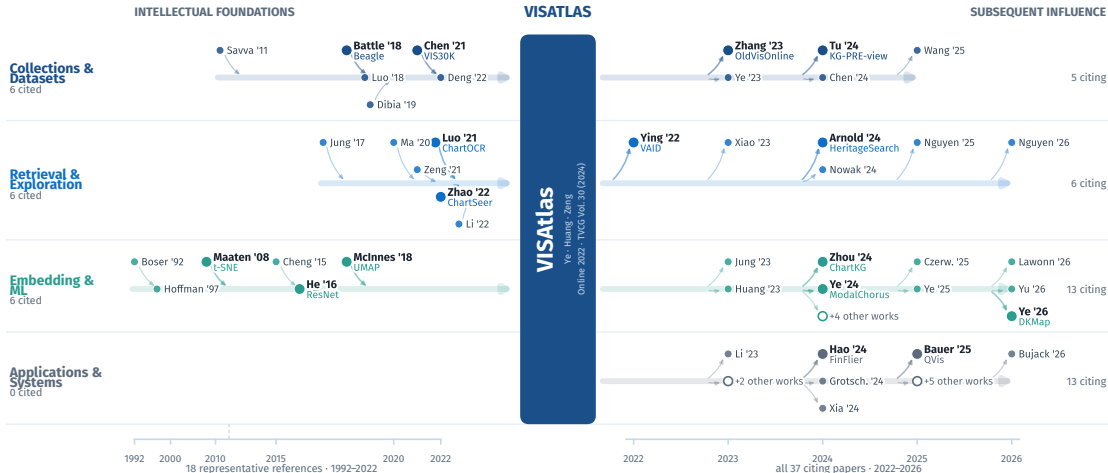
What visual features are prominent in a visualization?

What visual features are frequently composited in visualizations?”

VISAtlas represents the visual appearance of visualizations as interpretable embeddings, enabling users to explore, compare, and query visualization collections based on visual features rather than only extrinsic metadata.

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Related Work: Citation Provenance



Where did VISATlas come from intellectually? · Which research themes contributed to it? · Where did its influence propagate afterward?